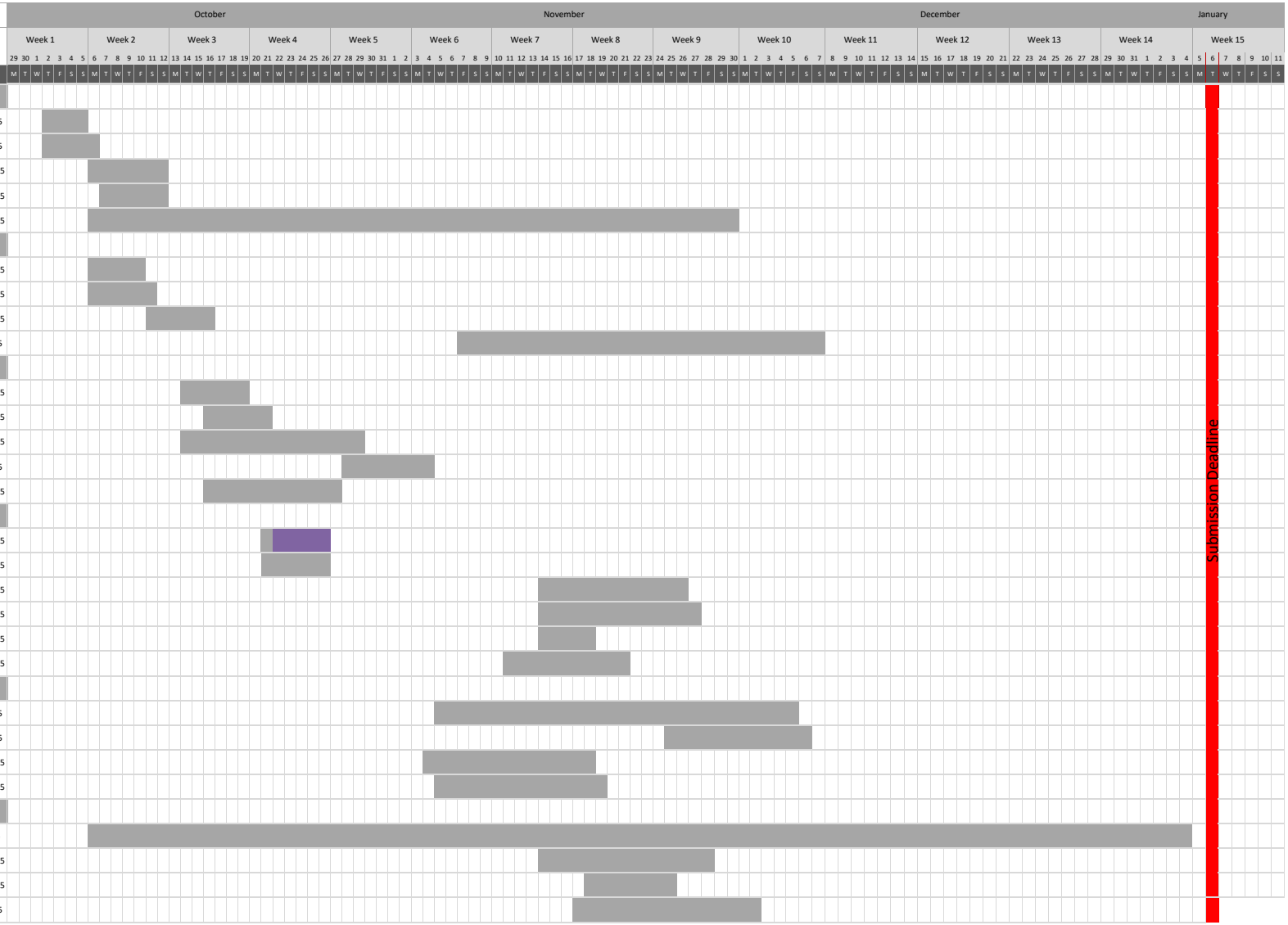
Group Design & Integration Project

University of the West of England

UFMFV8-15-3
Group 5I

Project Start: Thu, 02/10/2025

Display Project Timeline: 1

TASK	ASSIGNED TO	PROGRESS	START	END
Project Initial Planning				
Team formation and project selection	Group	100%	2/10/25	5/10/25
Create documentation and group communic	Logan	100%	2/10/25	6/10/25
Equipment rental	Logan	100%	6/10/25	12/10/25
Assign planning tasks	Group	100%	7/10/25	12/10/25
Start gantt chart	Omar	100%	6/10/25	30/11/25
Research & Requirments				
Define design constraints and scope	Group	100%	6/10/25	10/10/25
Complete stakeholder analysis	Akbar	100%	6/10/25	11/10/25
Hardware/software requirments and specs	Logan	100%	11/10/25	16/10/25
Circuit diagram creation	Akbar	100%	7/11/25	7/12/25
Design & Modelling				
Hardware/software requirments and specs	Logan	100%	14/10/25	19/10/25
Comparison between designs to reach final (	Jonny	100%	16/10/25	21/10/25
3D print test tube rack and grippers	Ben	100%	14/10/25	29/10/25
FOS for gripper and calculations	Ben	100%	28/10/25	4/11/25
Redesign grippers based on results	Ben	100%	16/10/25	27/10/25
Integration & Testing				
~~RaspberryPi testing and servo control valida~~	Logan	25%	21/10/25	26/10/25
ESP32 testing and servo control validation	Logan	100%	21/10/25	26/10/25
Physicall testing of full system	Group	100%	14/11/25	26/11/25
Test adjustments and wiring	Group	100%	14/11/25	27/11/25
Implementation Phase	Logan	100%	14/11/25	18/11/25
Testing the robot's capabilities against the sp	Group	100%	11/11/25	21/11/25
Verification & Validation (V&V)				
Create and complete V&V tables	Ben	100%	5/11/25	5/12/25
Perform validation testing	Group	100%	25/11/25	6/12/25
Assembly and soldering	Logan	100%	4/11/25	18/11/25
Flow chart and level designs	Logan/Omar/Akbar	100%	5/11/25	19/11/25
Milestones				
Write up report and system design	Group	100%	6/10/25	4/1/26
Initial Testing of full system	Group	100%	14/11/25	28/11/25
Debugging and optimization	Group	100%	18/11/25	25/11/25
Documentaion and presentation draft/prep	Group	100%	17/11/25	2/12/25